Value:
0: No delay.
1: Add 50ns delay. (Default value).

SDA_OUTPUT_DELAYED
SDA_OUTPUT_DELAYED attribute is used to add 50ns additional delay to the SDAO signal.

/* synthesis SDA_OUTPUT_DELAYED= value */

Value:
0: No delay (Default value).
1: Add 50ns delay.

I2C_FIFO_ENB

"I2C_FIFO_ENB" attribute is used to enable or disable FIFO

/* synthesis I2C_FIFO_ENB= [value] */

Value:
ENABLED : FIFO mode will be enabled.
DISABLED : FIFO mode will be disabled.

iCE40UP (iCE40 Ultra Plus) Hard Macros

This section describes the following dedicated hard macro primitives available in iCE40UP device.

SB_HFOSC
SB_LFOSC
SB_RGBA_DRV
SB_LEDDA_IP
SB_IO_OD
SB_I2C
SB_SPI
SB_MAC16
SB_SPRAM256KA
SB_IO_I3C

SB_*HFOSC*

SB_HFOSC primitive generates 48MHz nominal clock frequency within +/- 10% variation, with user programmable divider value of 1, 2, 4, and 8. The HFOSC can drive either the global clock network or fabric routes directly based on the clock network selection.

Ports

SB_HFOSC Ports		
Signal Name	**Direction**	**Description**
CLKHFPU	Input	Power up the HFOSC circuit. After power up, oscillator output will be stable after 100us. Active High.
CLKHFEN	Input	Enable the clock output. Enable should be low for the 100us power up period. Active High.
CLKHF	Output	HF Oscillator output.

Parameters

Parameter	Parameter Values.	Description.
CLKHF_DIV	"0b00"	Sets 48MHz HFOSC output.
	"0b01"	Sets 24MHz HFOSC output.
	"0b10"	Sets 12MHz HFOSC output.
	"0b11"	Sets 6MHz HFOSC output

Default Signal Values

The iCEcube2 software assigns the following signal value to unconnected port:
Input CLKHFEN: Logic “0”
Input CLKHFPU: Logic “0”

Clock Network Selection

By default the oscillator use one of the dedicated clock networks in the device to drive the elements. The user may configure the oscillator to use the fabric routes instead of global clock network using the synthesis attribute.

Synthesis Attributes

/* synthesis ROUTE_THROUGH_FABRIC = <value> */

Value:
0: Use dedicated clock network. Default option.
1: Use fabric routes.

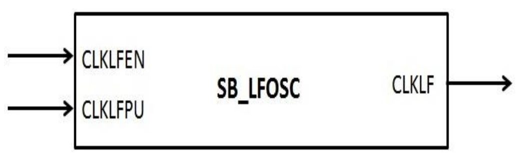
Verilog Instantiation

```
SB_HFOSC OSCInst0 (
.CLKHFEN(ENCLKHF),
.CLKHFPU(CLKHF_POWERUP),
.CLKHF(CLKHF)
) /* synthesis ROUTE_THROUGH_FABRIC= [0|1] */;

defparam OSCInst0.CLKHF_DIV = "0b00";
```

SB_LFOSC

SB_LFOSC primitive generates 10 KHz nominal clock frequency within +/- 10% variation. There is no divider on the LFOSC.
The LFOSC can drive either the global clock network or fabric routes directly based on the clock network selection



Ports

SB_LFOSC Ports		
Signal Name	**Direction**	**Description**
CLKLFPU	Input	Power up the LFOSC circuit. After power up, oscillator output will be stable after 100us. Active High.
CLKLFEN	Input	Enable the clock output. Enable should be low for the 100us power up period. Active High.
CLKLF	Output	LF Oscillator output.

Default Signal Values

The iCEcube2 software assigns the following signal value to unconnected port:
Input CLKLFEN: Logic "0"
Input CLKLFPU: Logic "0"

Clock Network Selection

By default the oscillator use one of the dedicated clock networks in the device to drive the elements. The user may configure the oscillator to use the fabric routes instead of global clock network using the synthesis attribute.

Synthesis Attributes

/* synthesis ROUTE_THROUGH_FABRIC = <value> */

Value:
0: Use dedicated clock network. Default option.
1: Use fabric routes.

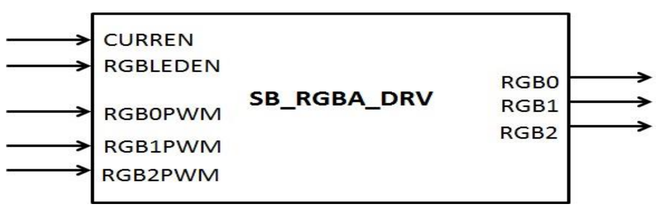
Verilog Instantiation

```
SB_LFOSC OSCInst1 (
       .CLKLFEN(ENCLKLF),
       .CLKLFPU(CLKLF_POWERUP),
       .CLKLF(CLKLF)
) /* synthesis ROUTE_THROUGH_FABRIC= [0|1] */;
```

SB_RGBA_DRV

SB_RGBA_DRV primitive is the RGB LED drive module which contains 3 dedicated open drain I/O pins for RGB LED outputs. Each of the RGB LED output is bonded out together with an SB_IO_OD primitive to the package pin. User can either use SB_RGB_DRV primitive or the SB_IO_OD primitive to drive the package pin, but not both.

The primitive allows configuration of each of the 3 RGB LED outputs individually. When the RGBx_CURRENT parameter of RGBx output is set to "0b000000", then SB_IO_OD can be used to drive the package pin.



Ports

SB_RGBA_DRV Ports		
Signal Name	**Direction**	**Description**
CURREN	Input	Enable the mixed signal control block to supply reference current to the IR drivers. When it is not enabled (CURREN=0), no current is supplied, and the IR drivers are powered down. Enabling the mixed signal control block takes 100us to reach a stable reference current value.
RGBLEDEN	Input	Enable the SB_RGB_DRV primitive. Active High.
RGB0PWM	Input	Input data to drive RGB0 LED pin. This input is usually driven from the SB_LEDD_IP.
RGB1PWM	Input	Input data to drive RGB1 LED pin. This input is usually

		driven from the SB_LEDD_IP.
RGB2PWM	Input	Input data to drive RGB2 LED pin. This input is usually driven from the SB_LEDD_IP.
RGB0	Output	RGB0 LED output.
RGB1	Output	RGB1 LED output.
RGB2	Output	RGB2 LED output.

Default Signal Values

The iCEcube2 software assigns the following signal value to unconnected port:
Input CURREN : Logic "0"
Input RGBLEDEN : Logic "0"
Input RGB0PWM : Logic "0"
Input RGB1PWM : Logic "0"
Input RGB2PWM : Logic "0"

Parameters

The SB_RGBA_DRV primitive contains the following parameter and their default values:

Parameter CURRENT_MODE = "0b0 ";
Parameter values:
"0b0" = Full Current Mode (Default).
"0b1" = Half Current Mode.

Parameter RGB0_CURRENT = "0b000000";
Parameter RGB1_CURRENT = "0b000000";
Parameter RGB2_CURRENT = "0b000000";
Parameter values:
"0b000000" = 0mA. // Set this value to use the associated SB_IO_OD instance at RGB // LED location.
"0b000001" = 4mA for Full Mode; 2mA for Half Mode
"0b000011" = 8mA for Full Mode; 4mA for Half Mode
"0b000111" = 12mA for Full Mode; 6mA for Half Mode.
"0b001111" = 16mA for Full Mode; 8mA for Half Mode
"0b011111" = 20mA for Full Mode; 10mA for Half Mode.
"0b111111" = 24mA for Full Mode; 12mA for Half Mode.

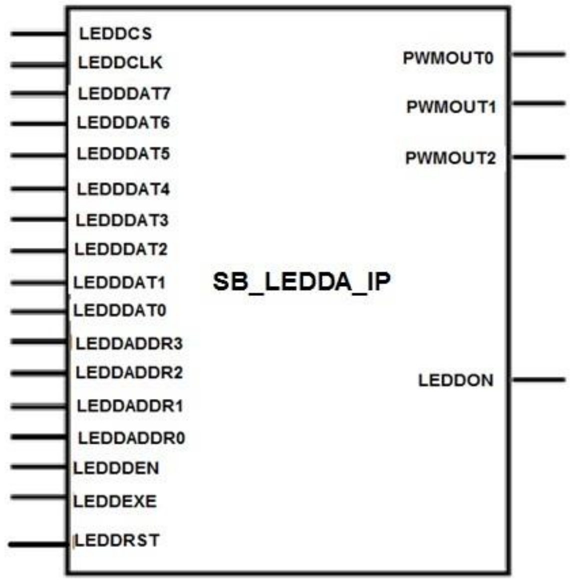
Verilog Instantiation

```
SB_RGBA_DRV RGBA_DRIVER (
      .CURREN(ENABLE_CURR),
      .RGBLEDEN(ENABLE_RGBDRV),
      .RGB0PWM(RGB0),
      .RGB1PWM(RGB1),
      .RGB2PWM(RGB2),
      .RGB0(LED0),
      .RGB1(LED1),
      .RGB2(LED2)
);

defparam RGBA_DRIVER.CURRENT_MODE = "0b0";
defparam RGBA_DRIVER.RGB0_CURRENT = "0b111111";
defparam RGBA_DRIVER.RGB1_CURRENT = "0b111111" ;
defparam RGBA_DRIVER.RGB2_CURRENT = "0b111111";
```

SB_LEDDA_IP

SB_LEDDA_IP primitive generates the RGB PWM outputs for the RGB LED drivers. The IP contains registers that are programmed in by the SCI bus interface signals.



Ports

SB_LEDDA_IP Ports		
Signal Name	**Direction**	**Description**
LEDDCS	Input	CS to write LEDD IP registers
LEDDCLK	Input	Clock to write LEDD IP registers
LEDDDAT7	Input	bit 7 data to write into the LEDD IP registers
LEDDDAT6	Input	bit 6 data to write into the LEDD IP registers
LEDDDAT5	Input	bit 5 data to write into the LEDD IP registers
LEDDDAT4	Input	bit 4 data to write into the LEDD IP registers
LEDDDAT3	Input	bit 3 data to write into the LEDD IP registers
LEDDDAT2	Input	bit 2 data to write into the LEDD IP registers
LEDDDAT1	Input	bit 1 data to write into the LEDD IP registers
LEDDDAT0	Input	bit 0 data to write into the LEDD IP registers
LEDDADDR3	Input	LEDD IP register address bit 3
LEDDADDR2	Input	LEDD IP register address bit 2
LEDDADDR1	Input	LEDD IP register address bit 1
LEDDADDR0	Input	LEDD IP register address bit 0
LEDDDEN	Input	Data enable input to indicate data and address are stable.
LEDDEXE	Input	Enable the IP to run the blinking sequence. When it is LOW, the sequence stops at the nearest OFF state.
LEDDRST	Input	Device level reset signal to reset all internal registers during the device configuration. This port is not accessible to user signals.
PWMOUT0	Output	PWM output 0
PWMOUT1	Output	PWM output 1
PWMOUT2	Output	PWM output 2

LEDDON	Output	Indicating the LED is on

Default Signal Values

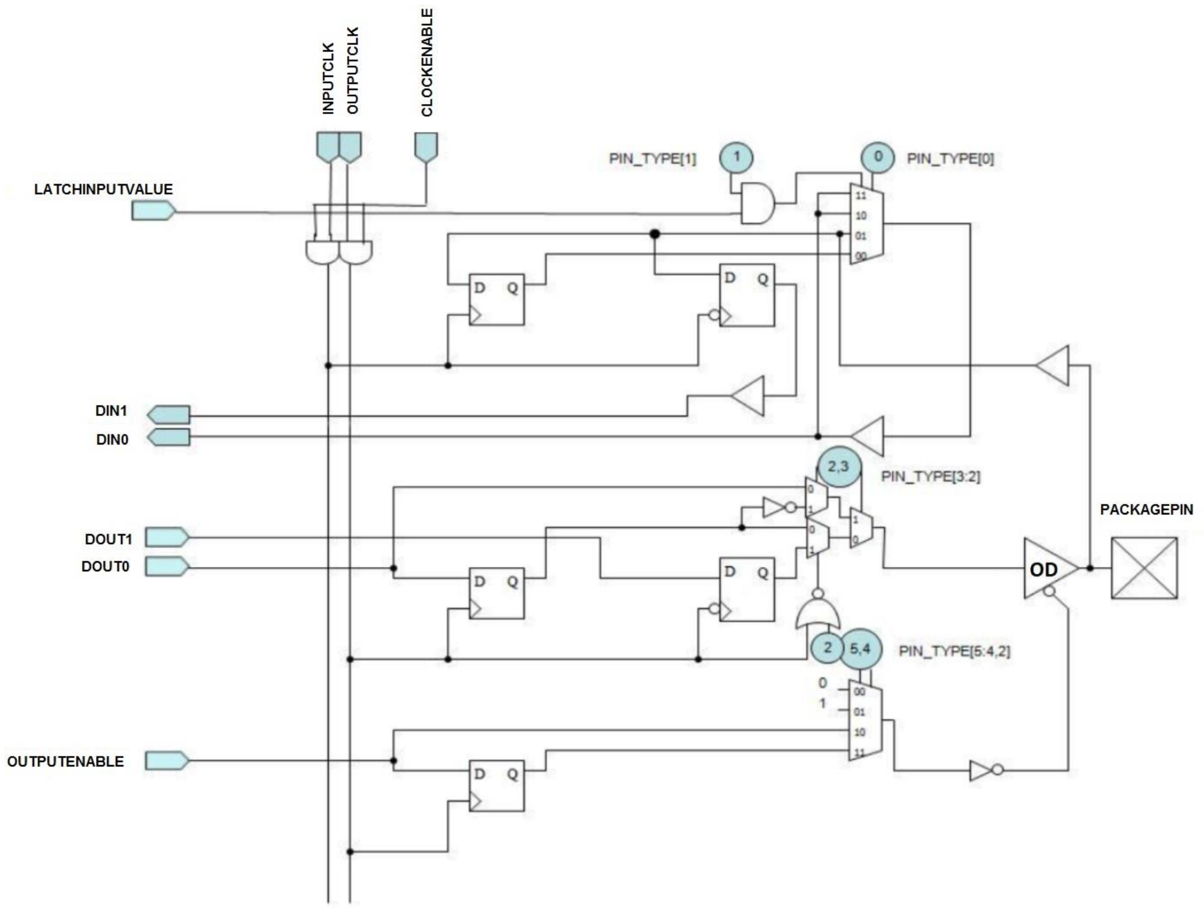
The iCEcube2 software assigns the logic “0” value to all unconnected input ports.

Verilog Instantiation

```
SB_LEDDA_IP  PWMgen_inst (
      .LEDDCS(ledd_cs),
      .LEDDCLK(led_clk),
      .LEDDDAT7(led_ip_data[7]),
      .LEDDDAT6(led_ip_data[6]),
      .LEDDDAT5(led_ip_data[5]),
      .LEDDDAT4(led_ip_data[4]),
      .LEDDDAT3(led_ip_data[3]),
      .LEDDDAT2(led_ip_data[2]),
      .LEDDDAT1(led_ip_data[1]),
      .LEDDDAT0(led_ip_data[0]),
      .LEDDADDR3(led_ip_addr[3]),
      .LEDDADDR2(led_ip_addr[2]),
      .LEDDADDR1(led_ip_addr[1]),
      .LEDDADDR0(led_ip_addr[0]),
      .LEDDDEN(led_ip_den),
      .LEDDEXE(led_ip_exe),
      .LEDDRST(led_ip_rst),
      .PWMOUT0(LED0),
      .PWMOUT1(LED1),
      .PWMOUT2(LED2),
      .LEDDON(led_on)
);
```

SB_IO_OD

The SB_IO_OD is the open drain IO primitive. When the Tristate output is enabled, the IO pulls down the package pin signal to zero. The following figure and Verilog template illustrate the complete user accessible logic diagram, and its Verilog instantiation.



Ports

SB_IO_OD Ports		
Signal Name	**Direction**	**Description**
PACKAGEPIN	Bidirectional	Bidirectional IO pin.
LATCHINPUTVALUE	Input	Latches/Holds the input data.
CLOCKENABLE	Input	Clock enable signal.
INPUTCLK	Input	Clock for the input registers.
OUTPUTCLK	Input	Clock for the output registers.
OUTPUTENABLE	Input	Enable Tristate output. Active high.
DOUT0	Input	Data to package pin.
DOUT1	Input	Data to package pin.
DIN0	Output	Data from package pin.
DIN1	Output	Data from package pin.

Default Signal Values

The iCEcube2 software assigns the logic “0” value to all unconnected input ports except for CLOCKENABLE.
Note that explicitly connecting logic “1” value to port CLOCKENABLE will result in a non-optimal implementation, since extra LUT will be used to generate the Logic “1”. If the user’s intention is to keep

the Input and Output registers always enabled, it is recommended that port CLOCKENABLE to be left unconnected.

Parameter Values

Parameter PIN_TYPE = 6'b000000;
// See Input and Output Pin Function Tables in SB_IO.
// Default value of PIN_TYPE = 6'b000000
Parameter NEG_TRIGGER = 1'b0;
// Specify the polarity of all FFs in the I/O to be falling edge when NEG_TRIGGER = 1. Default is 1'b0, rising edge.

Input and Output Pin Function Tables

Refer SB_IO Input and Output Pin Functional Table for the PIN_TYPE settings. Some of the output pin configurations are not applicable for SB_IO_OD primitive.

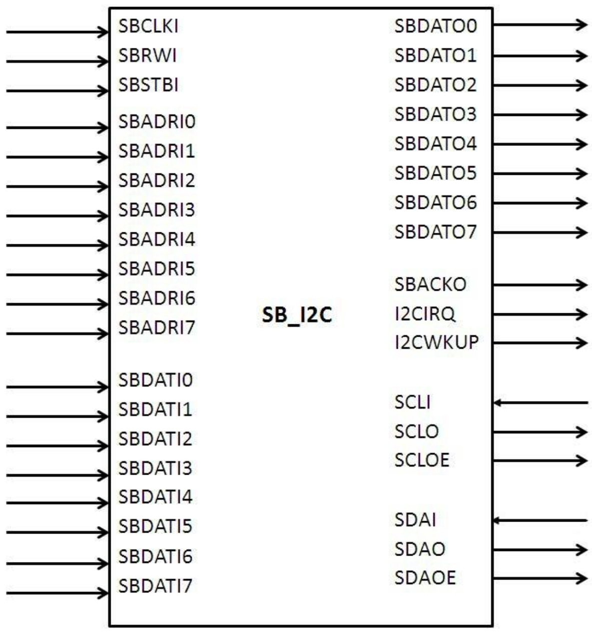
Verilog Instantiation

```
SB_IO_OD    OpenDrainInst0
(
.PACKAGEPIN (PackagePin),                 // User's Pin signal name
.LATCHINPUTVALUE (latchinputvalue),       // Latches/holds the Input value
.CLOCKENABLE (clockenable),               // Clock Enable common to input and
                                          // output clock
.INPUTCLK (inputclk),                     // Clock for the input registers
.OUTPUTCLK (outputclk),                   // Clock for the output registers
.OUTPUTENABLE (outputenable),             // Output Pin Tristate/Enable
                                          // control
.DOUT0 (dout0),                           // Data 0 - out to Pin/Rising clk
                                          // edge
.DOUT1 (dout1),                           // Data 1 - out to Pin/Falling clk
                                          // edge
.DIN0 (din0),                            // Data 0 - Pin input/Rising clk
                                          // edge
.DIN1 (din1)                              // Data 1 - Pin input/Falling clk
                                          // edge
);

defparam OpenDrainInst0.PIN_TYPE = 6'b000000;
defparam OpenDrainInst0.NEG_TRIGGER = 1'b0;
```

SB_I2C

iCE40UP device supports two I2C hard IP primitives , located at upper left corner and upper right corner of the chip.



Ports

The port interface is similar as iCE40LM/iCE5LP SB_I2C primitive. Refer Page 114.

Parameters

The parameters are same as ICE40LM/iCE5LP SB_I2C primitive.

Synthesis Attribute

I2C_CLK_DIVIDER

Synthesis attribute "I2C_CLK_DIVIDER" is used by PNR and STA tools for optimization and deriving the appropriate clock frequency at SCLO output with respect to the SBCLKI input clock frequency.

/* synthesis I2C_CLK_DIVIDER= Divide Value */

Divide Value: 0, 1, 2, 3 … 1023. Default is 0.

SDA_INPUT_DELAYED

SDA_INPUT_DELAYED attribute is used to add 50ns additional delay to the SDAI signal.

/* synthesis SDA_INPUT_DELAYED= value */

Value: